

[This question paper contains 8 printed pages.]

Your Roll No...2303

Sr. No. of Question Paper : 5594

J

Unique Paper Code : 2352012402

Name of the Paper : Multivariate Calculus

Name of the Course : B.A. / B.Sc. (H)

Semester : IV (DSC)

Duration : 3 Hours

Maximum Marks : 90

Instructions for Candidates

1. Write your Roll No. on the top immediately on receipt of this question paper.
2. Attempt all questions by selecting two parts from each question.
3. All questions carry equal marks.
4. Use of calculator is NOT allowed.

1. ~~(a)~~ Let $f(x,y) = \frac{\sin(x^2 + y^2)}{x^2 + y^2}$ for $(x,y) \neq (0,0)$. Find

the value of $f(0,0)$ for which $f(x,y)$ is continuous at $(0,0)$.

- ~~(b)~~ Compute the slope of the tangent line to the graph

of $f(x,y) = \frac{x^2 + y^2}{xy}$ at $P(1, -1, -2)$ in the direction

parallel to

(i) XZ plane

(ii) YZ plane.

- ~~(c)~~ Find $\frac{\partial w}{\partial r}$ where $w = e^{2x-y+3z^2}$ and $x = r + s - t$,

$y = 2r - 3s$, $z = \cos(rst)$.

2. (a) Let $f(x, y, z) = ye^{x+z} + ze^{y-x}$. At the point $P(2, 2, -2)$, find the unit vector pointing in the direction of most rapid increase of $f(x, y, z)$.

(b) Find all the critical points of $f(x, y) = (x-1)(y-1)(x+y-1)$ and classify each as a point of relative maximum, point of relative minimum or a saddle point.

(c) Find the minimum value of the function $f(x, y, z) = x^2 + y^2 + z^2$ subject to $4x^2 + 2y^2 + z^2 = 4$.

3. (a) (i) Find the volume of the solid bounded below by the rectangle R in the xy -plane and above by the graph of $z = 2x + 3y$; $R: 0 \leq x \leq 1, 0 \leq y \leq 2$.

(ii) Evaluate $\int_0^1 \int_x^1 e^{y^2} dy dx$.

(b) (i) Sketch the region of integration and write

an equivalent integral with the order of

integration reversed $\int_0^4 \int_{y/2}^{\sqrt{y}} f(x, y) dx dy.$

~~(ii)~~ Use a double integral for finding the volume of the solid region bounded above by the paraboloid $z = 6 - 2x^2 - 3y^2$ and below by the plane $z = 0$.

~~(c)~~ Use a double integral to find the area bounded by the curve $r = 1 + \sin \theta$.

4. ~~(a)~~ Use cylindrical co-ordinates to compute the integral

$$\iiint_D z(x^2 + y^2)^{-1/2} dx dy dz \text{ where } D \text{ is the solid}$$

bounded above by the plane $z = 2$ and bounded below by the surface $2z = x^2 + y^2$.

~~(b)~~ (i) Compute the iterated triple integral

$$\int_0^1 \int_0^y \int_0^{\ln y} e^{z+2x} dz dx dy.$$

(ii) Evaluate the iterated integral

$$\int_0^{\pi/2} \int_0^{\pi/4} \int_0^{\cos \phi} \rho^2 \sin \phi \, d\rho d\theta d\phi.$$

(c) Evaluate $\iint_R e^{(2y-x)/(y+2x)} dA$

where R is the trapezoid with vertices $(0,2)$, $(1,0)$, $(4,0)$ and $(0,8)$.

5. (a) A wire has the shape of the curve

$$x = \sqrt{2} \sin t \quad y = \cos t \quad z = \cos t \quad \text{for } 0 \leq t \leq \pi$$

If the wire has density $\delta(x, y, z) = xyz$ at each point (x, y, z) , what is its mass?

(b) Show that the vector field

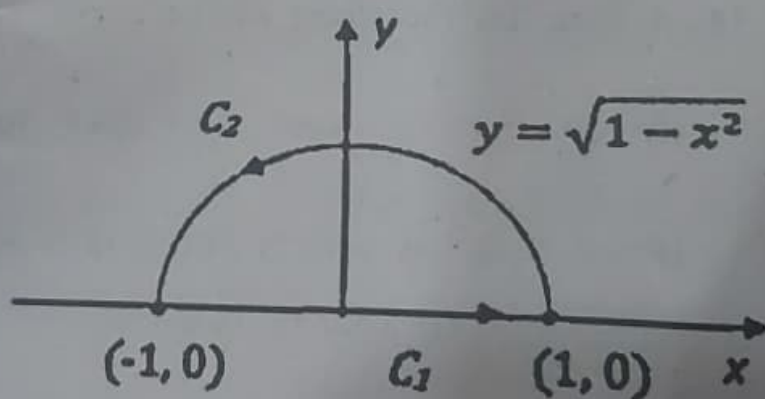
$$F = (e^x \sin y - y)\hat{i} + (e^x \cos y - x - 2)\hat{j}$$

is conservative and hence find a scalar potential function f for F .

(c) Verify Green's theorem for the line integral

$$\oint_C (-y \, dx + x \, dy)$$

where C is the closed path shown in the figure below



Path C

6. (a) Evaluate the surface integral

$$\iint_S g \, dS$$

where $g(x,y,z) = xz + 2x^2 - 3xy$ and S is that portion of the plane $2x - 3y + z = 6$ that lies over the unit square $R: 2 \leq x \leq 3, 2 \leq y \leq 3$.

- (b) Evaluate

$$\oint_C \left(\frac{1}{2} y^2 \, dx + z \, dy + x \, dz \right)$$

where C is the curve of intersection of the plane $x + z = 1$ and the ellipsoid $x^2 + 2y^2 + z^2 = 1$, oriented counterclockwise.

- (c) Use the divergence theorem to evaluate



$$\iiint_S \mathbf{F} \cdot \mathbf{N} \, dS$$

Where $F = (x^5 + 10x y^2 z^2) \hat{i} + (y^5 + 10y x^2 z^2) \hat{j} + (z^5 + 10z x^2 y^2) \hat{k}$, and S is the closed hemispherical surface $z = \sqrt{1 - x^2 - y^2}$ together with the disc $x^2 + y^2 \leq 1$ in xy -plane and N is the outward unit normal vector field.